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DSC 640-T301 Data Presentation & Visual

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Assignment 7 & 8:

We are going to explore an age-old question: [The jackpot is a lie?](#)

Audience:

The intended audience for this project is the New York State Lottery's executive and analytics team. They are very familiar with the terminology from the data and information from the lottery. They are also responsible for evaluating transparency and fairness within the lottery operations. A second audience included the general public, who may be interested in the overall findings. This project is designed to balance clarity and analytical accuracy, ensuring anyone can easily understand the materials.

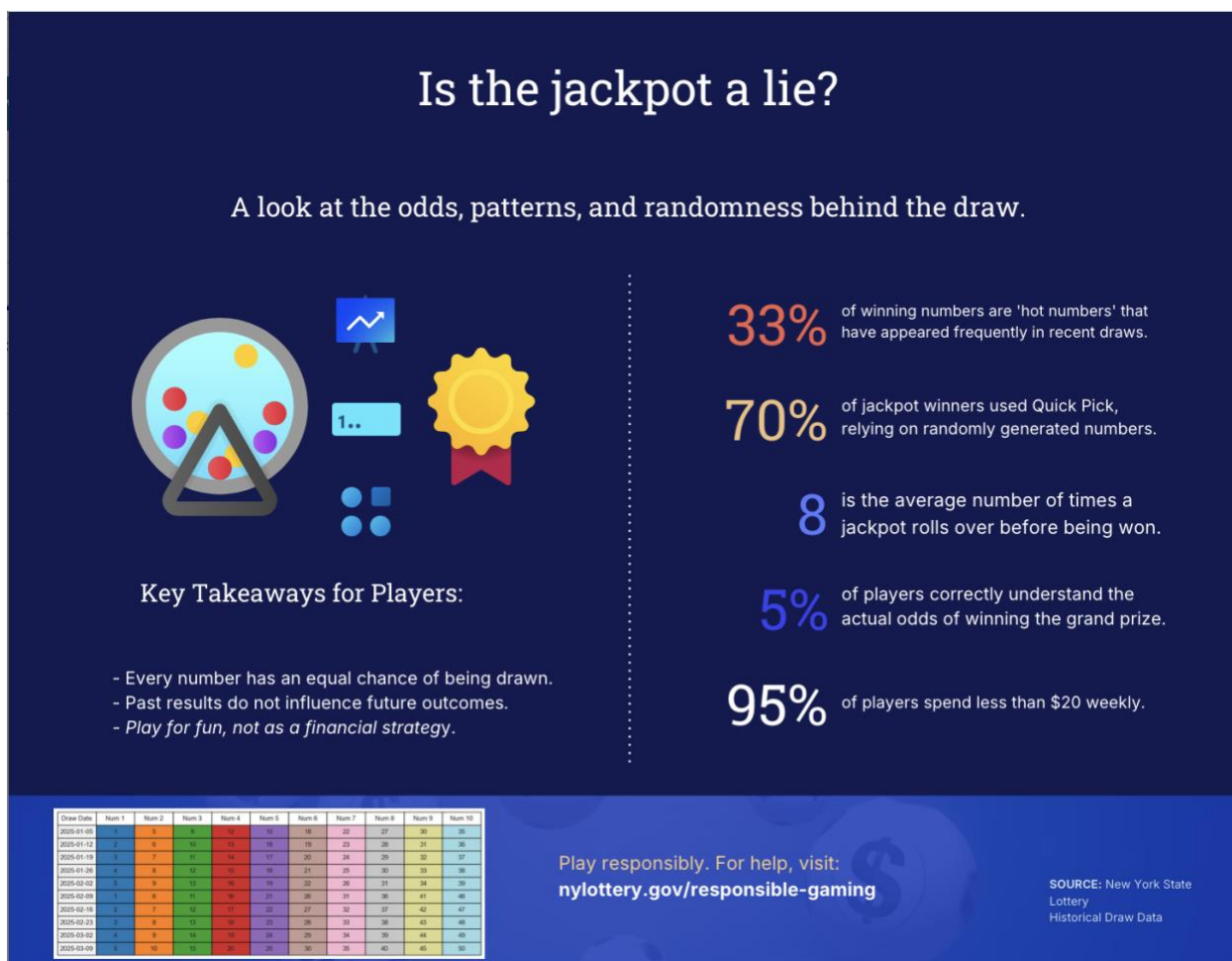
Purpose:

The purpose of this project is to analyze historical lottery data to determine whether winning numbers occur randomly or show identifiable patterns. By examining results from Mega Millions, Powerball, and Pick 10, the study can represent the fairness and transparency of lottery operations. The goal is to provide insights that support data-driven decision-making and promote greater public trust. The finding will encourage the New York State Lottery to enhance transparency and reinforce responsible gaming practices.

Call to Action:

The New York State Lottery aims to promote transparency, build public trust, and encourage responsible gaming without relying on complex tools or dashboards. To achieve this, the Lottery should provide brief educational notes explaining odds and randomness, helping players understand how the game really works. Using the available data, the Lottery is encouraged to share a clear infographic that summarizes historical draw patterns, number frequencies, and rollover information.

Infographic:



I used Venngage to make the infographic. <https://infograph.venngage.com/infographics>

Medium for Infographic:

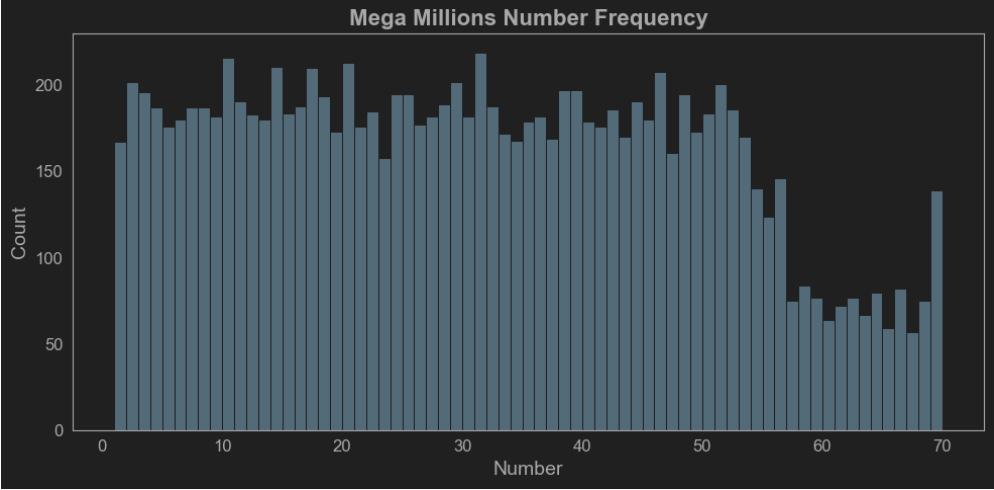
The findings from all three datasets will be presented through a single-page infographic that has been designed to be visually clear and easy to understand by both Lottery stakeholders and the general public. The infographic will summarize key insights from the historical draw data, including number frequencies, patterns, and rollover information, along with a brief explanation of odds and randomness. This medium is simple, accessible, and effective for communicating transparency and promoting responsible gaming.

Design for Infographic:

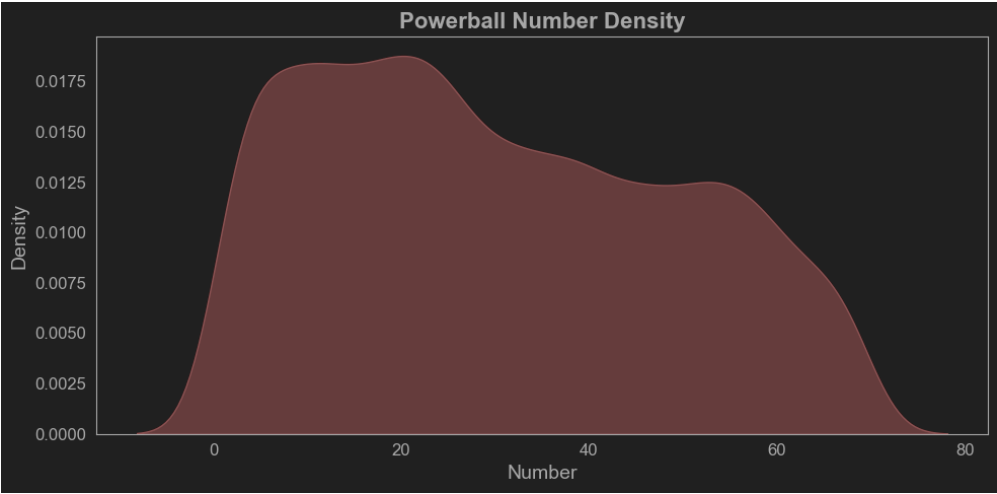
The infographic was designed using Venngage, which helps make infographics and a variety of presentation materials. I also added in my table to the bottom, along with the website to where to play and my resource to the dataset, which is important for transparency. I put 5 large statistical stats and 3 general / fun bullet points to keep the reader engaged. Trying to pull them back to the call of action. Then adding in little fun visuals, is for people who aren't interested facts. The title, is the title of the project, Is the jackpot a lie? I kept it simple, with a dark and light blue background and then a few little colorful pictures that remind you of the lottery.

Visuals:

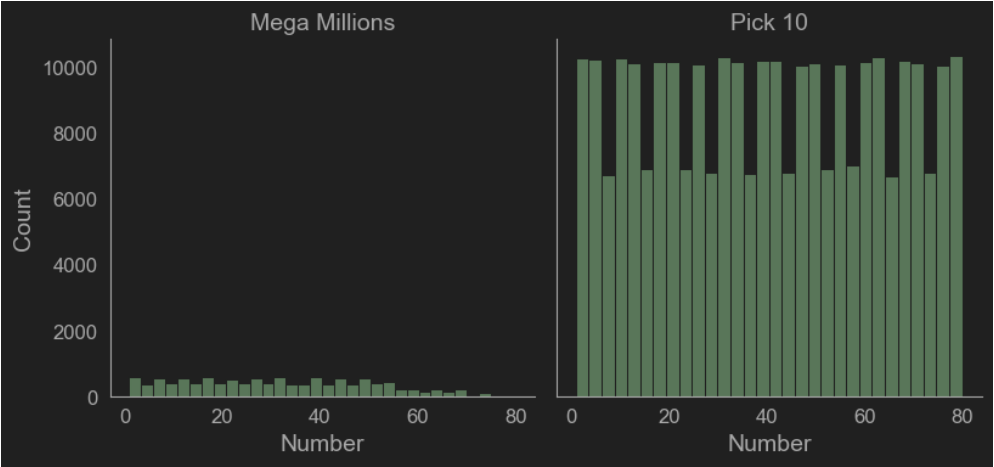
Graph 1: Histogram - Mega Millions Frequency



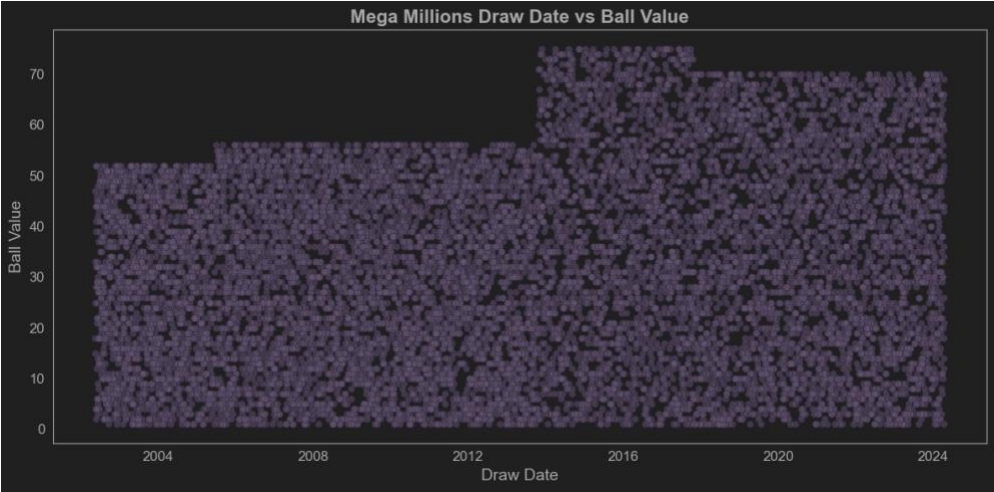
Graph 2: Density Plot – Powerball



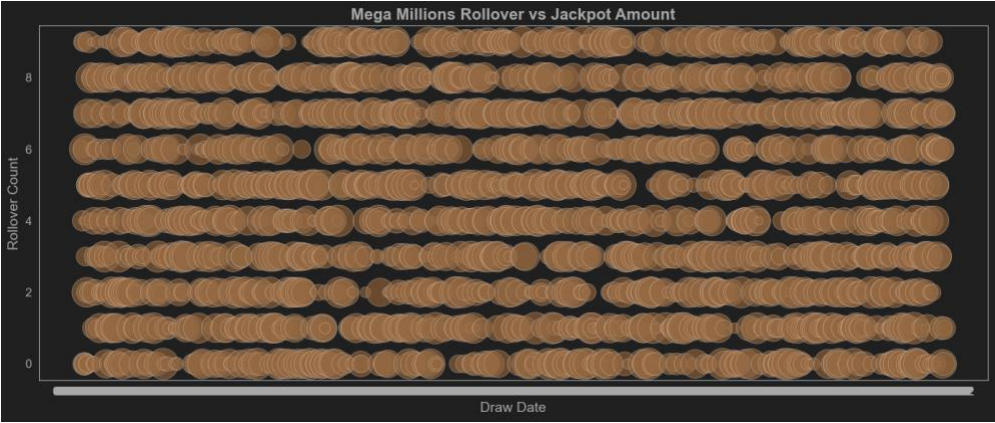
Graph 3: Small Multiples - Frequency per Game



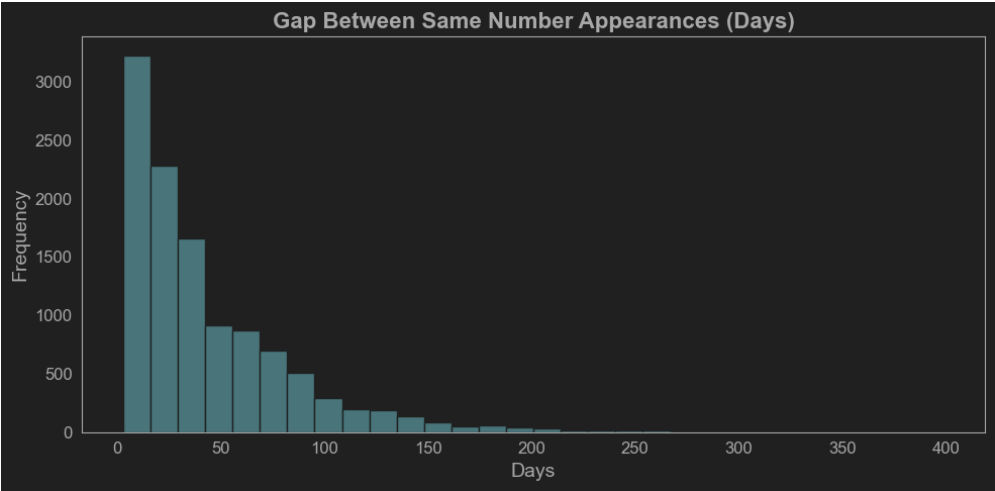
Graph 4: Scatterplot - Draw Date vs Ball Value (Mega Millions)



Graph 5: Bubble Chart - Rollover vs Jackpot Amount



Graph 6: Histogram - Gap between same number appearances (Mega Millions)



Graph 7: Pick 10 Table

Draw Date	Num 1	Num 2	Num 3	Num 4	Num 5	Num 6	Num 7	Num 8	Num 9	Num 10
2025-01-05	1	5	9	12	15	18	22	27	30	35
2025-01-12	2	6	10	13	16	19	23	28	31	36
2025-01-19	3	7	11	14	17	20	24	29	32	37
2025-01-26	4	8	12	15	18	21	25	30	33	38
2025-02-02	5	9	13	16	19	22	26	31	34	39
2025-02-09	1	6	11	16	21	26	31	36	41	46
2025-02-16	2	7	12	17	22	27	32	37	42	47
2025-02-23	3	8	13	18	23	28	33	38	43	48
2025-03-02	4	9	14	19	24	29	34	39	44	49
2025-03-09	5	10	15	20	25	30	35	40	45	50

Ethical Considerations:

All the data used in the study was sourced from the official New York open-data portals, ensuring credibility and transparency. The analysis respects privacy and does not involve any personally identifiable information. Visualizations and insights were designed to promote understanding without encouraging gambling behavior, supporting responsible gaming practices.

Conclusion:

The study of Mega Millions, Powerball, and Pick 10 historical data shows that while some numbers appear slightly more often, overall patterns are consistent with randomness. No strategy can reliably predict winning numbers.

The infographic and visualizations provide a clear, easy-to-understand summary for both lottery staff and the public. Sharing these insights supports transparency, builds trust, and encourages responsible gaming. As the lottery remains to be a game of chance and the jackpot is not predictable, which confirms that winning is possible, it is also random.

Resources:

[Vennage](#)

Datasets:

[Lottery Mega Millions Winning Numbers Beginning 2002](#)

[Lottery Pick 10 Winning Numbers Beginning 1987](#)

Lottery Powerball Winning Numbers Beginning 2010

Dataset Source:

data.ny.gov